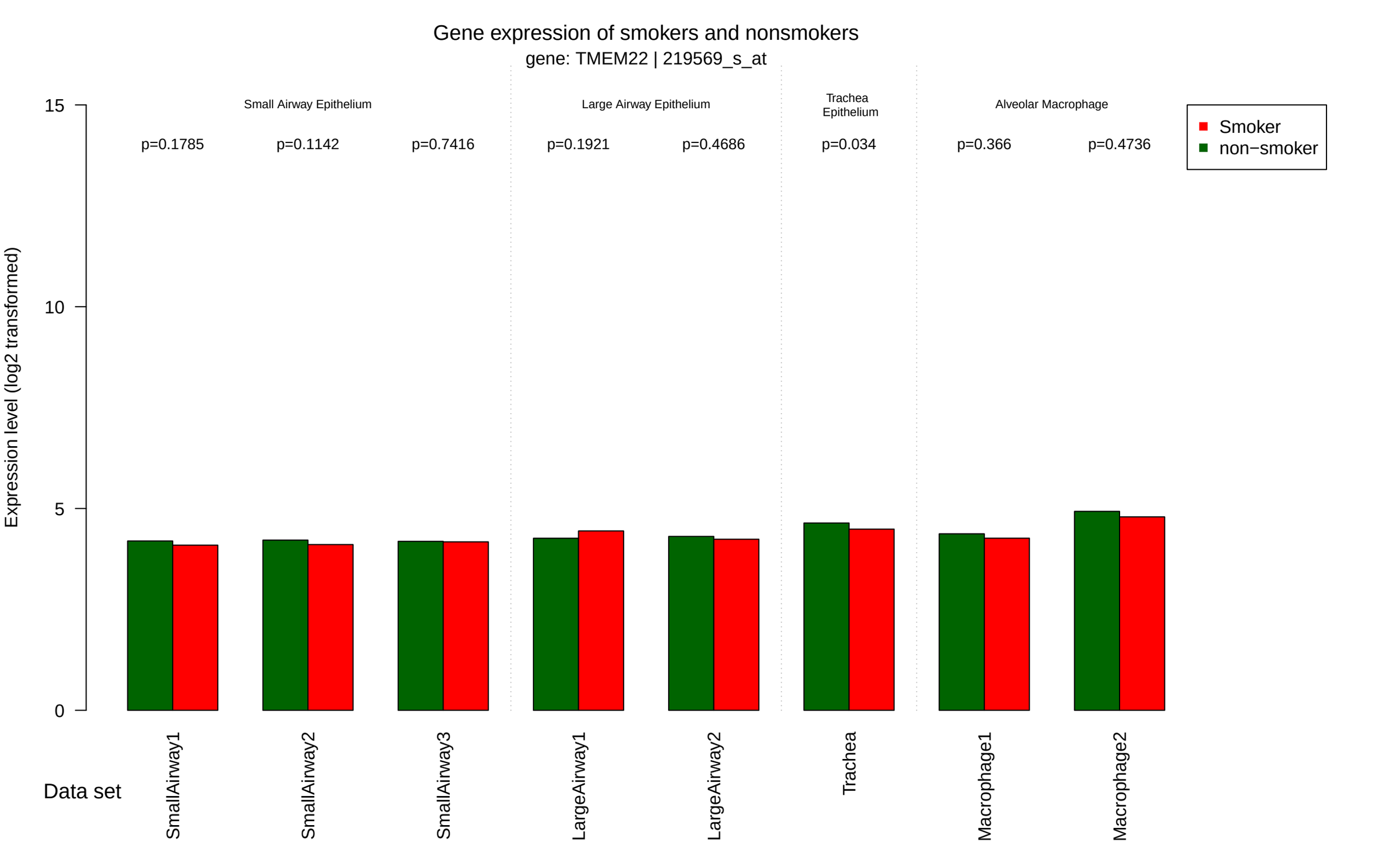
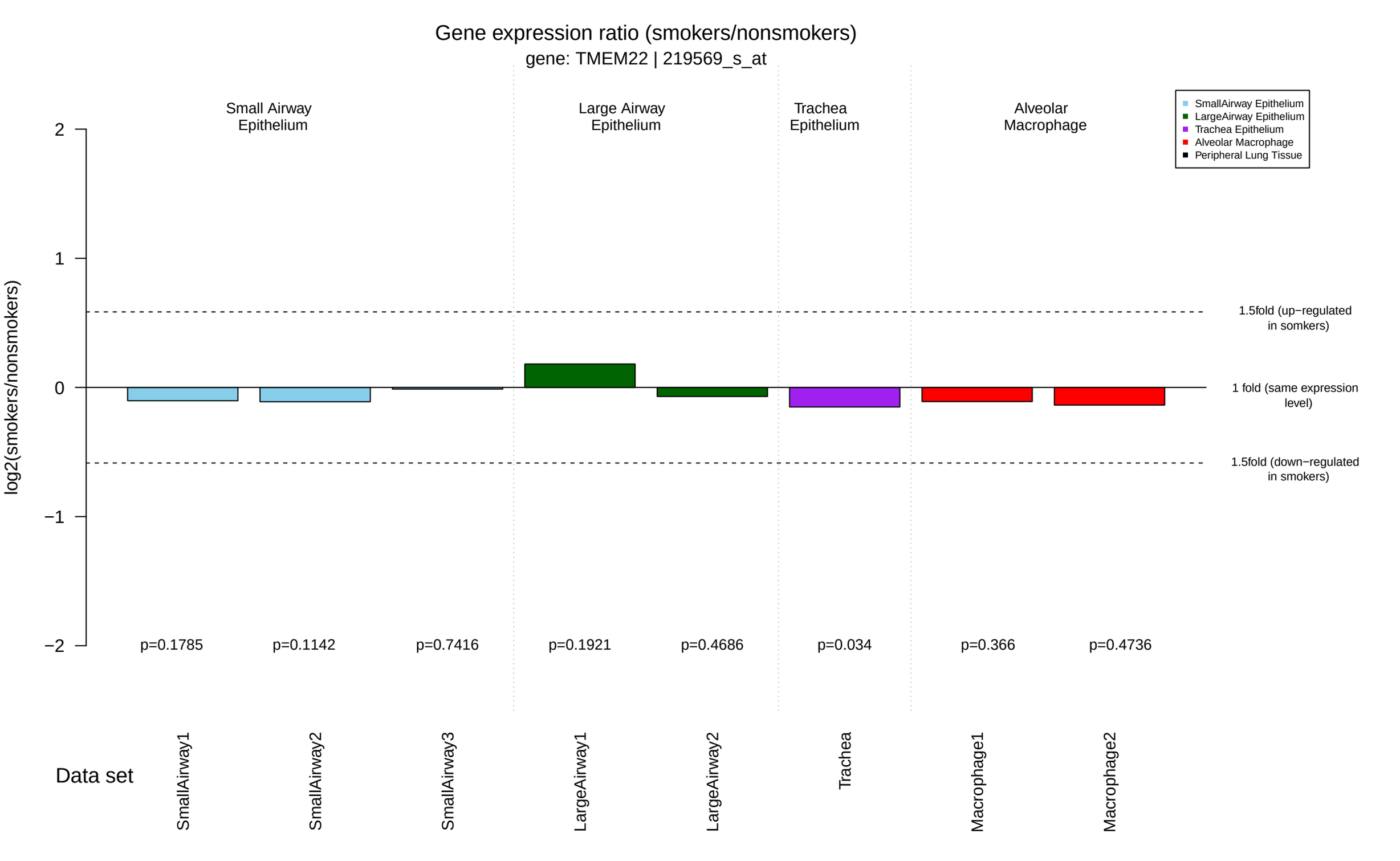
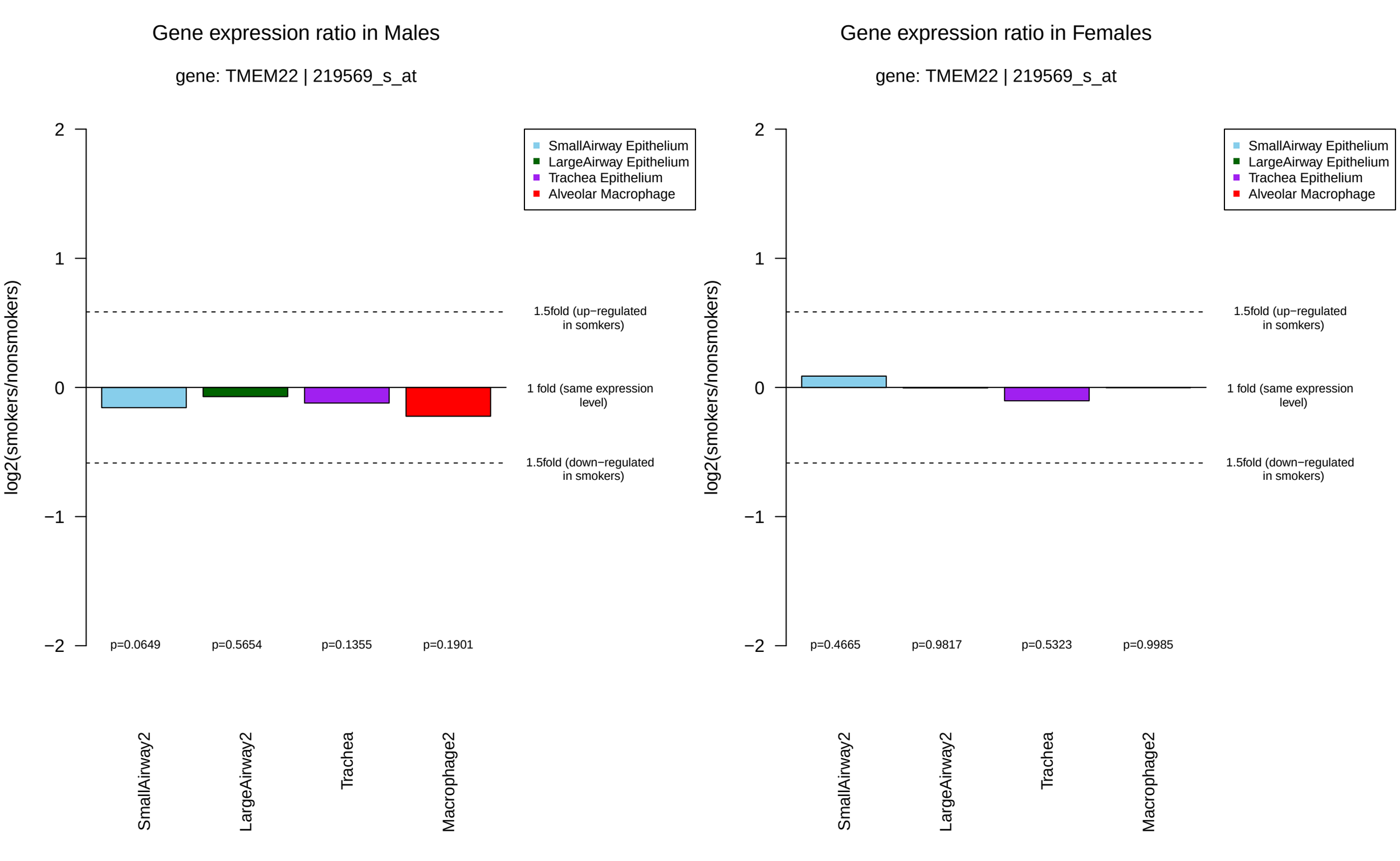
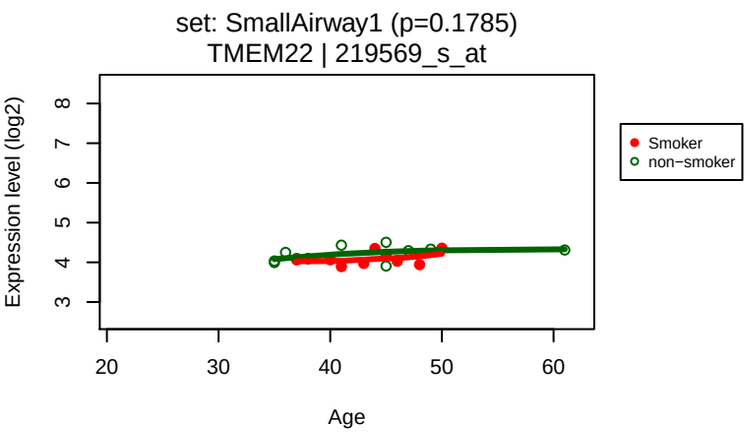
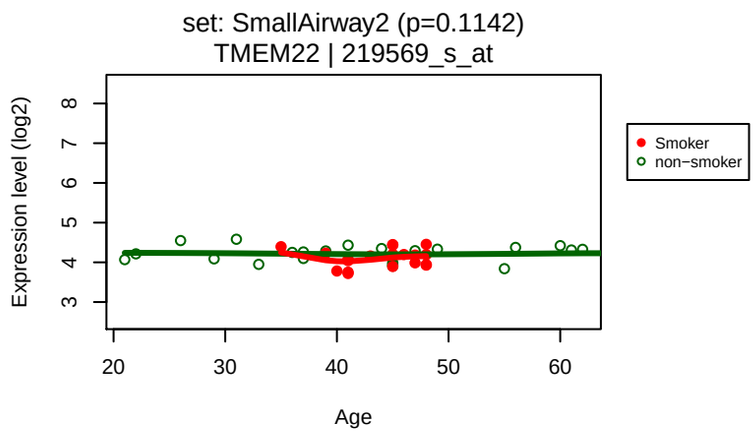
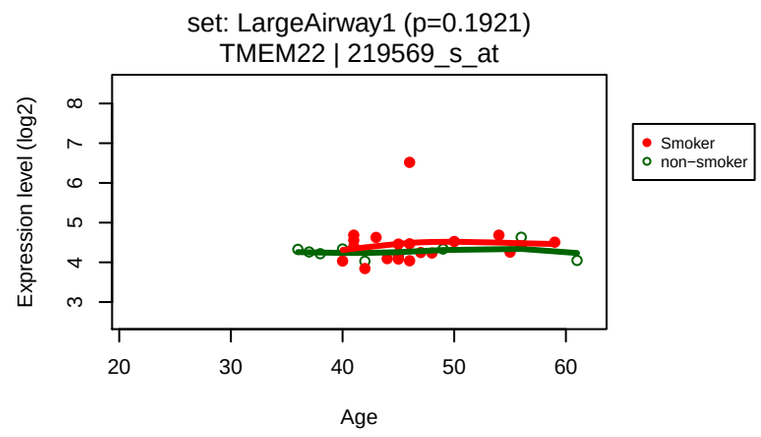

Gene expression of smokers and nonsmokers
gene: TMEM22 | 219569_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1785
p=0.1142
p=0.7416
p=0.1921
p=0.4686
p=0.034
p=0.366
p=0.4736
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

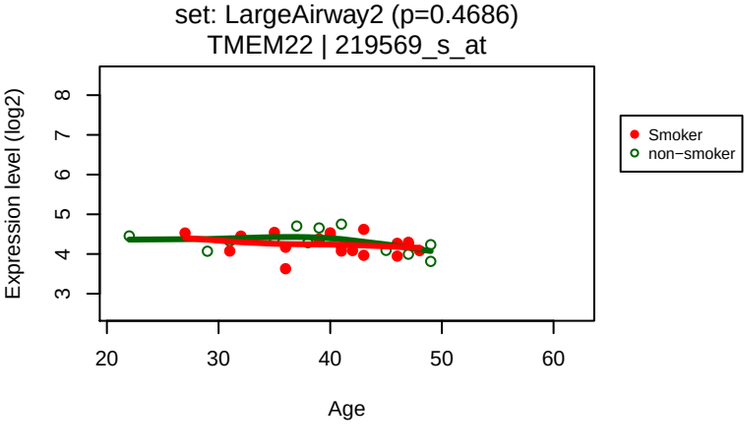

Gene expression ratio (smokers/nonsmokers)
gene: TMEM22 | 219569_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1785
p=0.1142
p=0.7416
p=0.1921
p=0.4686
p=0.034
p=0.366
p=0.4736
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

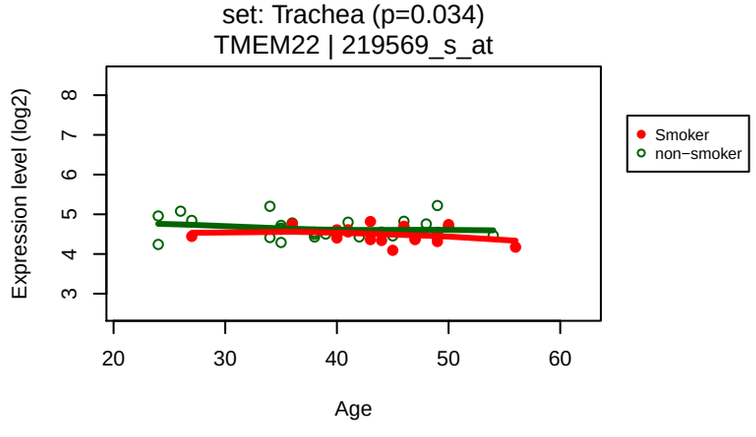

Gene expression ratio in Males
gene: TMEM22 | 219569_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0649
p=0.5654
p=0.1355
p=0.1901
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM22 | 219569_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4665
p=0.9817
p=0.5323
p=0.9985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

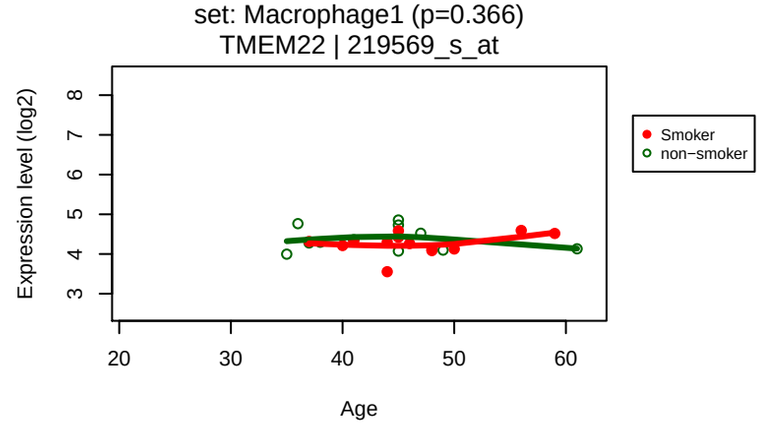

set: SmallAirway1 (p=0.1785)
TMEM22 | 219569_s_at
Expression level (log2)
Age
Smoker
non−smoker

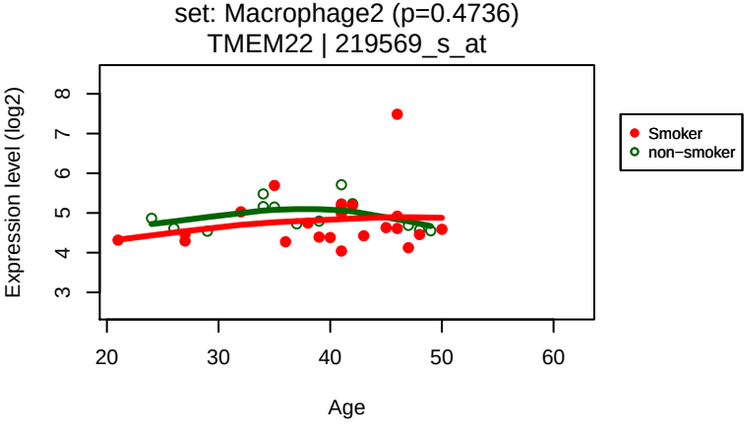

set: SmallAirway2 (p=0.1142)
TMEM22 | 219569_s_at
Expression level (log2)
Age
Smoker
non−smoker

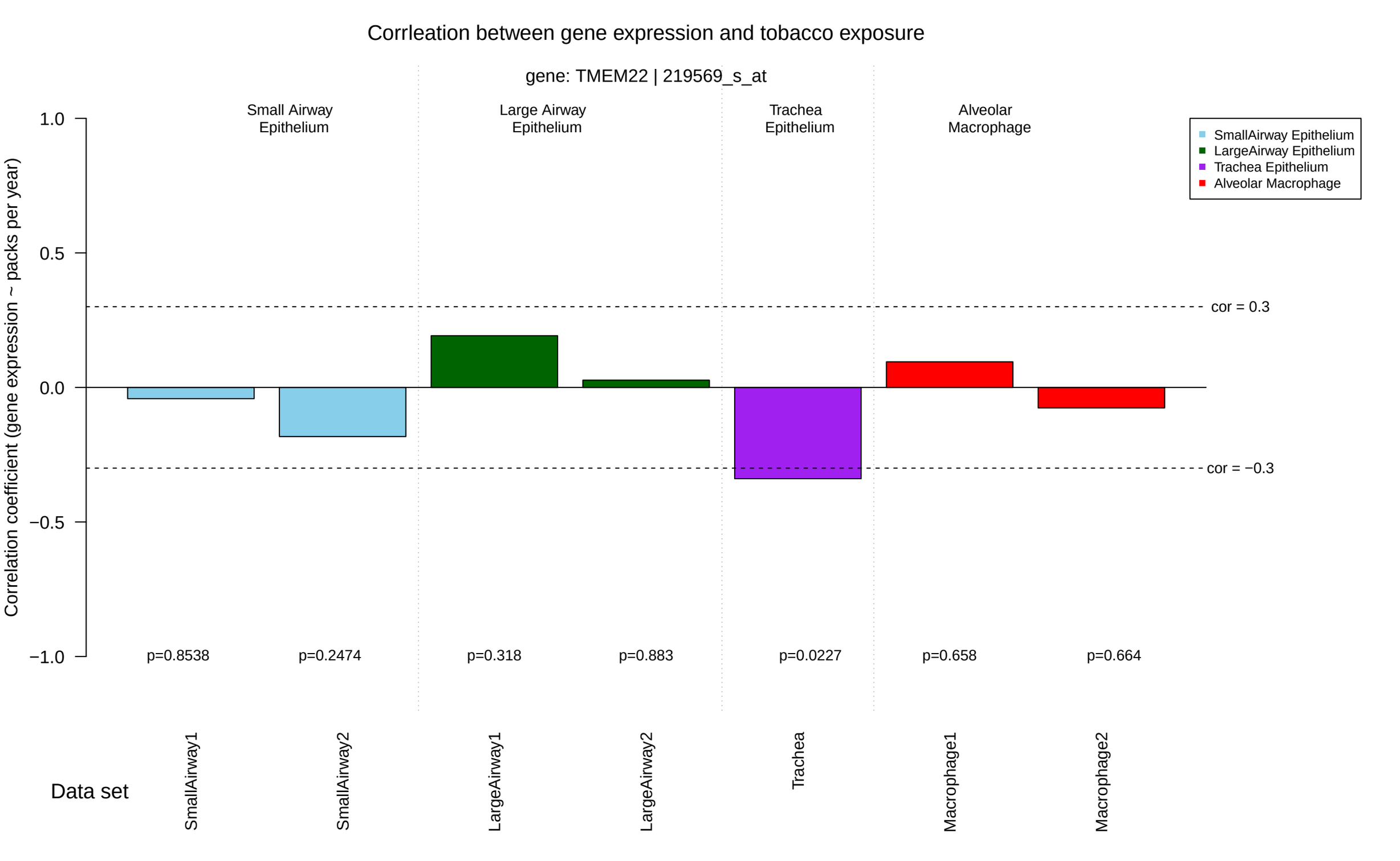

set: LargeAirway1 (p=0.1921)
TMEM22 | 219569_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4686)
TMEM22 | 219569_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.034)
TMEM22 | 219569_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.366)
TMEM22 | 219569_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4736)
TMEM22 | 219569_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM22 | 219569_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8538
p=0.2474
p=0.318
p=0.883
p=0.0227
p=0.658
p=0.664
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2